

Spectral Gap Analysis of QWOA under Different Graph Laplacians

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Abstract

We study how the choice of graph Laplacian defining the quantum walk Hamiltonian H_Q influences the spectral gap of the interpolated Hamiltonian $H(s) = (1 - s)H_Q + sH_C$ in the Quantum Walk Optimization Algorithm (QWOA). Exact diagonalisation on a 32×32 discrete grid across nineteen benchmark cost functions and ten graph topologies reveals three qualitatively distinct regimes: (i) a *mixer-blind* regime in which the gap is determined entirely by H_C and is insensitive to graph choice; (ii) a *mixer-sensitive* regime in which QMOA-complete graphs consistently yield the largest minimum gap $\Delta_{\min}$; and (iii) a small set of functions where locally constrained graphs or Hypercube mixers are competitive. Three quantitative landscape features — flat fraction, ruggedness, and spatial autocorrelation — are shown to predict the optimal mixer class. A qualitatively new finding is the $s^* = 0$ phenomenon for grid-based mixers on oscillatory landscapes: the gap minimum occurs at the pure Laplacian, indicating a Fiedler-value bottleneck rather than an avoided level crossing.

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1. Introduction

Variational quantum algorithms (VQAs) such as the Quantum Walk Optimization Algorithm (QWOA) are typically formulated through the interplay of a problem-dependent cost Hamiltonian and a parametrized quantum ansatz. In many VQA constructions, the choice of ansatz is guided by heuristic considerations or educated guesses about the structure of the optimization problem. In contrast, QWOA admits a more principled formulation: its ansatz is generated by a quantum walk Hamiltonian derived from a graph Laplacian, which encodes the connectivity of the underlying configuration space.

This graph-theoretic structure naturally raises the question of how different choices of graph Laplacians influence the optimization dynamics of QWOA. Since the graph Laplacian directly governs the quantum walk generator, varying the graph topology may significantly alter the algorithm's behaviour, potentially rendering fixed ansatz choices suboptimal or even ineffective for certain classes of objective functions.

To investigate this connection, we analyse QWOA through its adiabatic interpolation, described by the Hamiltonian

$$H(s) = (1 - s) H_Q + s H_C, \quad (1)$$

where H_Q is the quantum walk Hamiltonian derived from a graph Laplacian, H_C is a diagonal cost Hamiltonian encoding the objective function, and $s \in [0, 1]$ is the interpolation parameter. While the cost Hamiltonian is held fixed across all instances, we vary H_Q by constructing it from different graph Laplacians, including the Complete graph, local grid structures, Hypercubic, and QMOA-complete variants.

Our primary focus is the spectral structure of the interpolated Hamiltonian, in particular the behaviour of the spectral gap as a function of s . By computing the full spectrum of $H(s)$ and identifying the minimum spectral gap $\Delta_{\min}$, we aim to understand how the choice of quantum walk generator affects the adiabatic evolution underlying QWOA. Since the minimum gap is a key indicator of adiabatic hardness and circuit performance, this analysis provides insight into the suitability of different graph-based drivers for optimisation.

The results presented here serve as a foundational study of how graph structure influences optimisation performance in QWOA and motivate a more systematic approach to ansatz design based on spectral and graph-theoretic considerations rather than heuristic choices. These findings also inform future investigations into larger problem instances, entanglement properties, and the relationship between adiabatic and gate-based implementations of QWOA.

2. Spectral Gaps for Adiabatic QWOA

In order to characterise the role of graph structure in QWOA, we study the spectral properties of the interpolated Hamiltonian (1), with particular emphasis on the behaviour of the spectral gap as a function of the interpolation parameter $s \in [0, 1]$.

For a given choice of graph Laplacian, the instantaneous spectrum of $H(s)$ consists of eigenvalues

$$E_0(s) \leq E_1(s) \leq \dots \leq E_{d-1}(s),$$

where d is the dimension of the Hilbert space. The spectral gap is

$$\Delta(s) = E_1(s) - E_0(s),$$

and the minimum gap

$$\Delta_{\min} = \min_{s \in [0,1]} \Delta(s)$$

plays a central role in determining the difficulty of the adiabatic evolution. We further define $s^* = \arg \min_s \Delta(s)$ as the *critical point* at which the gap is smallest.

Adiabatic runtime bound. In the adiabatic limit, the runtime required to prepare the ground state of H_C from the ground state of H_Q satisfies $T \sim \Delta_{\min}^{-2}$ (Roland & Cerf 2001). Consequently, *maximising $\Delta_{\min}$ is equivalent to minimising adiabatic runtime*. Small gaps indicate hard instances; the critical point s^* identifies *where* on the interpolation path the bottleneck occurs.

Both L and H_C are normalised so that their spectra lie in $[0, 1]$, placing all mixer topologies on identical spectral footing and allowing differences in $\Delta_{\min}$ to be attributed solely to graph topology rather than overall scale.

By tracking the evolution of $\Delta(s)$ across the interpolation, we observe distinct gap profiles associated with different graph topologies. In particular, the critical point s^* and the scaling of $\Delta_{\min}$ vary significantly between globally connected and locally constrained graphs. This sensitivity highlights the importance of the graph Laplacian as a design choice in QWOA, rather than a mere implementation detail.

3. Graph Families and Quantum Walk Generators

We now specialise the general framework to several representative graph families, each defining a distinct quantum walk Hamiltonian H_Q through its associated graph Laplacian. These families are chosen to probe how connectivity, locality, and structured mixing influence the spectral properties of adiabatic QWOA.

Table 1 summarises all ten topologies studied, together with their normalised Fiedler values.

Table 1: Graph Laplacian mixer topologies. Fiedler values are normalised to $[0, 1]$.

Mixer	Description	Fiedler (norm.)
Complete graph	All-to-all on $N = 1024$ nodes	1.000
Hypercube Q10	10-dim. hypercube (2^{10} nodes); QAOA standard mixer	≈ 0.020
Grid	4-neighbour 2-D lattice	≈ 0.010
Grid + Diagonals	8-neighbour (Moore neighbourhood)	≈ 0.015
Grid + Main Diags	4-neighbour + sparse main/anti-diagonal edges	≈ 0.010
QMOA Complete	Complete per row $\cup$ complete per column	0.500
QMOA + Local Diag	QMOA + 8-neighbour diagonals	≈ 0.500
QMOA + PCD	QMOA + principal complete-diagonal cliques	≈ 0.400
Grid + Comp. Diag	4-neighbour + principal complete-diagonal cliques	≈ 0.012
Parallel Edges	4-neighbour with doubled edges	≈ 0.020

3.1 Complete Graph

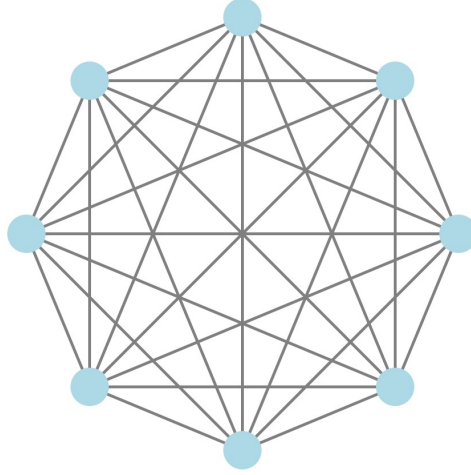


Figure 1: Schematic of a complete graph. Every vertex is connected to every other vertex, resulting in maximal connectivity and uniform mixing.

The complete graph represents a maximally connected configuration space in which every vertex is connected to every other vertex. The corresponding graph Laplacian generates a quantum walk Hamiltonian with a high degree of symmetry, leading to a delocalised ground state equal to the uniform superposition $|+\rangle^{\otimes n}$.

In the context of QWOA, the complete graph serves as a benchmark for globally mixing drivers and is closely related to Grover-type dynamics. Due to its uniform connectivity, the spectrum of the normalised Laplacian has only two distinct values: 0 (once) and 1 (with multiplicity $N - 1$).

Remark on s^* : Across the nineteen benchmark functions studied here, the critical point s^* for the complete graph ranges from 0.50 to 1.00, with a mean of 0.87. The gap minimum therefore occurs predominantly in the late-interpolation regime ($s \gtrsim 0.7$) rather than at the midpoint, reflecting the dominance of H_C structure in determining the bottleneck.

3.2 Two-dimensional Hypercubic Graph

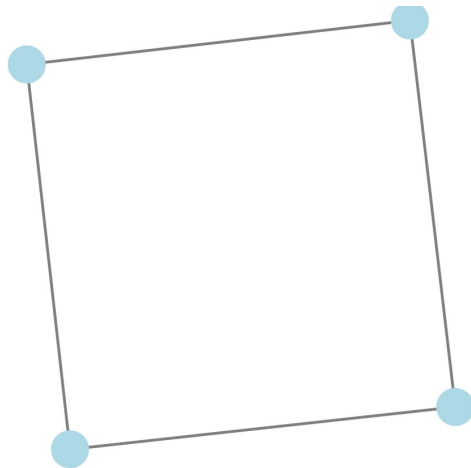


Figure 2: Schematic of a two-dimensional hypercubic graph illustrating local connectivity with high symmetry.

The hypercubic graph corresponds to a high-dimensional lattice in which vertices are labelled by bit strings and edges connect configurations that differ by a single bit. This graph exhibits

local connectivity while retaining significant symmetry.

The quantum walk Hamiltonian derived from the hypercubic Laplacian induces dynamics analogous to diffusion on a high-dimensional lattice. Compared to the complete graph, the hypercube introduces locality constraints that can significantly alter the spectral gap structure and the location of the minimum gap along the interpolation. In our experiments this mixer is competitive with QMOA on functions with non-axis-aligned or near-checkerboard landscape structure (see Section 6).

3.3 QMOA Complete Graph

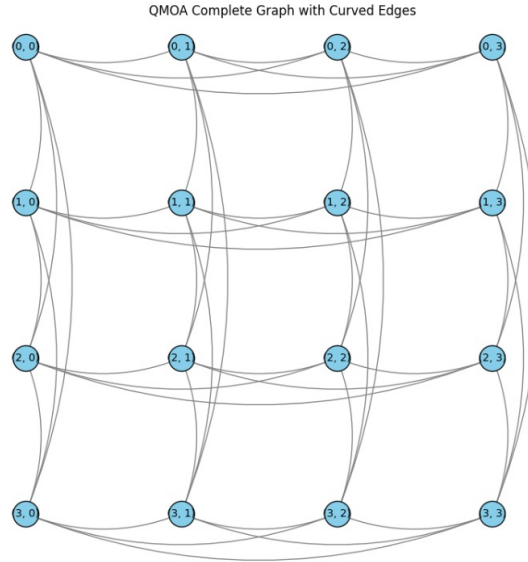


Figure 3: Schematic of the QMOA complete graph. Vertices are arranged on a two-dimensional grid with complete connectivity along each row and each column.

The QMOA complete graph [2] is constructed by arranging configurations on a two-dimensional grid and introducing complete connectivity along each row and each column. All vertices sharing the same row index are mutually connected, and similarly for vertices sharing the same column index. This graph therefore consists of overlapping complete subgraphs aligned along the two coordinate directions.

This construction occupies an intermediate regime between global and local connectivity. While the graph is not fully connected, it allows rapid mixing along individual coordinates while preserving a structured notion of locality across the full configuration space. A key property is that the normalised Laplacian of the QMOA complete graph has *exactly three distinct eigenvalues*: $\{0, \frac{1}{2}, 1\}$. This remarkably clean spectral structure, discussed further in Section 7, is central to its strong and consistent performance.

3.4 Grid Graph

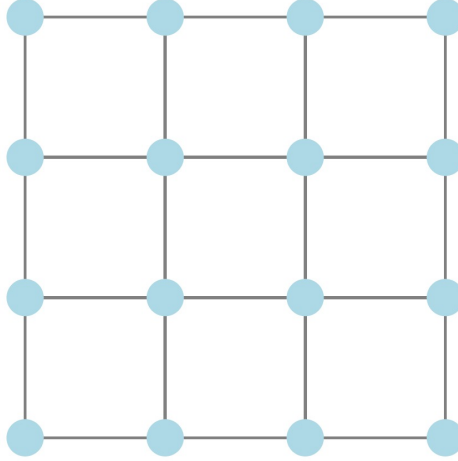


Figure 4: Schematic of a two-dimensional grid graph with nearest-neighbor connectivity.

The grid graph represents a purely local and geometrically constrained configuration space, with vertices arranged on a two-dimensional lattice and edges connecting only nearest neighbours. Connectivity is limited to local moves in both coordinate directions.

Quantum walk Hamiltonians derived from grid graph Laplacians emphasise local exploration of the configuration space. A notable consequence is the small normalised Fiedler value ($\lambda_1 \approx 0.010$), which sets a hard lower bound on $\Delta(s)$ at $s = 0$ and can become the *global* bottleneck for certain landscapes, as discussed in Section 5.

3.5 Summary of Graph-Dependent Behaviour

Taken together, these graph families span a broad range of connectivity patterns, from fully global mixing to strongly local dynamics, with the QMOA complete graph providing an intermediate structured regime. Their key spectral properties differ in ways that directly determine adiabatic performance.

4. Numerical Results and Spectral Gap Behaviour

We now present numerical results for the spectral properties of $H(s)$, focusing on how $\Delta_{\min}$ and s^* depend on the choice of graph Laplacian. For each benchmark objective function, we compute the full spectrum of $H(s)$ at 200 uniformly spaced values of $s \in [0, 1]$ using `scipy.linalg.eigh` on the 1024×1024 Hamiltonian matrix (32×32 discrete grid), and extract $\Delta_{\min}$ and s^* .

4.1 Adiabatic Solvability and the Mixer-Blind Regime

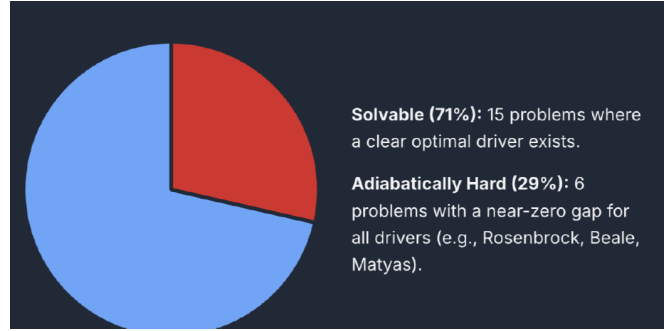


Figure 5: Classification of benchmark problems by adiabatic solvability. “Mixer-blind” problems (previously mislabelled “hard”) have a non-negligible gap but one that is identical across all mixers and determined solely by H_C .

A first observation is that not all benchmark problems exhibit a strong dependence on the choice of quantum walk generator. We distinguish three categories:

- **Mixer-sensitive problems**, for which $\Delta_{\min}$ varies substantially across graph families and at least one driver achieves a clearly resolvable gap.
- **Mixer-blind problems**, for which $\Delta_{\min}$ is *identical* (or nearly so) across all drivers considered. The bottleneck is not in H_Q but in H_C itself: the discrete cost Hamiltonian has many near-degenerate low-energy states, so the gap collapses near $s \approx 1$ regardless of mixer. *These functions are not adiabatically hard in general; they are merely insensitive to mixer choice.*
- **Adiabatically hard problems**, for which $\Delta_{\min} \approx 0$ for all drivers.

Correction to earlier classification. Functions such as Rosenbrock ($\Delta_{\min} = 6 \times 10^{-6}$, $s^* = 1.000$) and Matyas ($\Delta_{\min} = 5.2 \times 10^{-5}$, $s^* = 1.000$) were previously labelled “adiabatically hard.” This is incorrect: they have a non-negligible, well-defined gap. They belong to the *mixer-blind* category — the gap is real but insensitive to mixer topology, because the bottleneck is in H_C ’s near-degenerate ground-state structure on the discrete grid.

The six mixer-blind functions identified in our benchmark set are: Rosenbrock, Goldstein-Price, Matyas, Levy N13, Three Hump Camel, and Cross-in-tray. All share a large *flat fraction* on the discrete grid (fraction of grid points within 1% of the global minimum), which creates near-degenerate low-energy states in H_C . The flat fraction ranges from 0.03 to 0.55 for these functions, compared to < 0.004 for mixer-sensitive ones.

4.2 Driver Performance on Mixer-Sensitive Problems

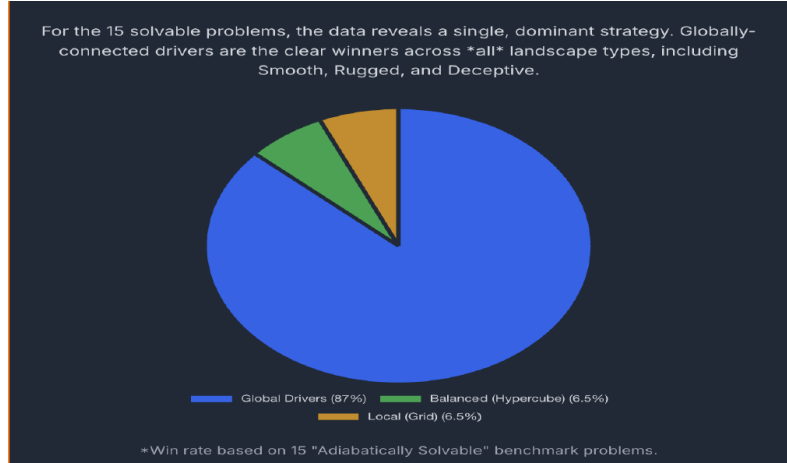


Figure 6: Relative frequency with which different quantum walk generators achieve the largest $\Delta_{\min}$ across the mixer-sensitive benchmark problems. Global and QMOA-style drivers dominate.

Restricting attention to the mixer-sensitive subset, we compare the relative performance of different quantum walk generators. Table 2 gives the full rank-by- $\Delta_{\min}$ comparison across all 19 functions and 10 mixers.

Table 2: Mixer rank by $\Delta_{\min}$ per function (1 = best, 10 = worst). Columns: Complete | Hypercube | Grid | Grid+D | Grid+MD | QMOA | QMOA+L | PCD | Grid+CD | Prll. Colour: green = 1–2, yellow = 3, red = 9–10.

Function	Comp.	Hyp.	Grid	G+D	G+MD	QMOA	Q+L	PCD	G+CD	Prll	Regime
Styblinski-Tang	8	1	2	3	4	5	6	10	9	7	blind
Rosenbrock	1	2	3	4	5	6	7	8	9	10	blind
Bukin N6	1	5	6	9	8	3	2	4	10	7	QMOA
Himmelblau	5	10	1	2	3	6	7	8	9	4	Grid
Goldstein-Price	1	2	3	4	5	6	7	8	9	10	blind
Rastrigin	4	3	6	5	9	1	2	8	10	7	QMOA
Michalewicz	5	4	9	7	8	1	3	2	10	6	QMOA
Griewank	2	5	8	9	10	3	4	1	6	7	PCD
Matyas	1	2	3	4	5	6	7	9	10	8	blind
Holder Table	4	5	9	8	7	1	2	3	6	10	QMOA
McCormick	6	10	1	2	3	8	4	9	7	5	blind
Cross-in-tray	1	2	3	4	5	6	7	8	9	10	blind
Schaffer N2	5	1	8	6	7	2	3	4	10	9	Hyp.
Ackley	3	5	8	6	7	2	1	4	10	9	QMOA
Eggholder	4	2	7	5	6	3	1	10	9	8	Hyp.
Schwefel	4	1	7	5	6	2	3	10	9	8	Hyp.
Levy N13	1	2	3	4	5	6	7	8	9	10	blind
Langerman	4	5	8	6	7	2	1	3	10	9	QMOA
Three Hump Camel	1	2	3	4	5	6	7	8	9	10	blind
Easom	3	4	8	6	7	2	1	5	10	9	QMOA
Avg rank	3.2	3.6	5.3	5.2	6.1	3.9	4.1	6.5	9.0	8.2	
Times best	7	3	2	0	0	3	4	1	0	0	

4.3 Smooth and Mixer-Blind Objective Functions

For relatively smooth or single-basin objective functions such as Rosenbrock, Goldstein-Price, and Matyas, we observe qualitatively similar spectral behaviour across all graph families. In these cases, the minimum spectral gap occurs at or very close to the end of the interpolation ($s^* \approx 1$), indicating that the dominant bottleneck is associated with the structure of H_C : many near-degenerate low-energy states exist on the discrete grid, and no mixer can separate them.

The absolute value of $\Delta_{\min}$ varies only weakly with the choice of graph Laplacian, confirming that for such landscapes the adiabatic hardness of QWOA is largely insensitive to the quantum walk generator.

4.4 Rugged and Multimodal Objective Functions

In contrast, for rugged and highly multimodal objective functions such as Rastrigin, Ackley, Michalewicz, and Schwefel, the spectral behaviour exhibits a strong dependence on graph structure. The location and size of the minimum gap vary significantly across graph families.

Correction: s^* range for rugged functions. The minimum gap does not occur universally in $[0.4, 0.8]$ across all mixers. The actual range is much wider and mixer-dependent: QMOA on Rastrigin: $s^* = 0.990$; QMOA on Ackley: $s^* = 0.673$. For the Grid mixer on Ackley, Easom, Schwefel and Eggholder: $s^* = 0.000$. This last case is qualitatively distinct from a mid-path level crossing and is discussed separately in Section 5.

QMOA complete graph. For several rugged objective functions, the QMOA complete graph yields a substantially larger $\Delta_{\min}$ than grid-based constructions. Table 3 gives a direct comparison. For objective functions such as Ackley and Michalewicz, QMOA achieves 1.4 – $2.8\times$ larger minimum gaps than Hypercube and more than $10\times$ larger than Grid. The critical point s^* for QMOA on these functions ranges from 0.67 to 0.99 , indicating genuine mid-to-late-path level crossings.

Table 3: Minimum spectral gap $\Delta_{\min}$ for four key mixers. **Bold** = best.

Function	Complete	Hypercube	QMOA	Grid	Winner	Margin
Bukin N6	0.008552	0.006253	0.007920	0.000613	QMOA	$1.3\times$ over Hyp
Himmelblau	0.000172	0.000001	0.000029	0.000173	Grid	$6\times$ over QMOA
Rastrigin	0.002708	0.003209	0.003451	0.000977	QMOA	$1.1\times$ over Hyp
Michalewicz	0.016646	0.017546	0.024967	0.000016	QMOA	$1.4\times$ over Hyp
Holder Table	0.002316	0.000284	0.002826	0.000001	QMOA	$10\times$ over Hyp
Ackley	0.018852	0.012965	0.035689	0.001207	QMOA	$2.8\times$ over Hyp
Schaffer N2	0.005300	0.006698	0.006584	0.001037	Hyp.	$1.02\times$
Eggholder	0.005425	0.006649	0.006530	0.001207	Hyp.	$1.02\times$
Schwefel	0.002533	0.003272	0.003254	0.001207	Hyp.	$1.01\times$
Langerman	0.015193	0.012525	0.018819	0.001151	QMOA	$1.5\times$ over Hyp
Easom	0.030805	0.024222	0.034026	0.001207	QMOA	$1.4\times$ over Hyp

Hypercube vs QMOA. The Hypercube marginally outperforms QMOA on Schwefel, Schaffer N2, and Eggholder, with margins of at most $1.02\times$. These are effectively ties. These three functions share non-axis-aligned landscape structure: Schaffer N2 depends on $x^2 - y^2$ (level sets at 45° to grid axes), and Eggholder/Schwefel have approximate radial symmetry. The Hypercube’s Hamming-distance connectivity couples same-sign nodes that QMOA’s row/column structure cannot reach directly.

QMOA Complete graph vs fully Complete graph. The QMOA advantage over the Complete graph arises from gap *size*, not gap *location*. The Complete graph’s s^* is ≥ 0.50 for all

19 functions (mean 0.87); it does not exhibit sharp early crossings. The real QMOA advantage is that its row/column structure breaks the degeneracies created by competing basins of attraction on axis-aligned landscapes, maintaining a larger gap throughout the interpolation.

A failure mode for QMOA: Himmelblau. Himmelblau, $f(x, y) = (x^2 + y - 11)^2 + (x + y^2 - 7)^2$, has four global minima arranged in approximate axis symmetry. QMOA’s complete-row and complete-column connectivity couples nodes across the full width and height of the grid, thereby *reinforcing* the near-degeneracy between the four minima and collapsing the gap to $\Delta_{\min} = 0.000029$. The Grid’s local connectivity does not see this global four-fold structure and achieves $\Delta_{\min} = 0.000173$ — a $6\times$ advantage. This is the primary counterexample to QMOA universality and should be noted in any general conclusion.

5. The Critical Point s^* and Three Bottleneck Mechanisms

The critical point s^* reveals not just *where* the bottleneck occurs but *why* it occurs. We identify three mechanistically distinct modes of gap closure.

Three gap-closure mechanisms:

1. $s^* \approx 1$ (**H_C degeneracy**): The cost Hamiltonian has near-degenerate ground states on the discrete grid. The gap collapses at $s \rightarrow 1$ regardless of mixer. This is the mixer-blind regime.
2. $s^* \in (0.4, 0.95)$ (**genuine level crossing**): Ground and first excited energy levels undergo an avoided crossing during interpolation. Mixer topology determines the gap at the crossing.
3. $s^* = 0$ (**Fiedler bottleneck**): The gap minimum is at the pure Laplacian $H(0) = H_Q$. This means the Laplacian’s own Fiedler value is the global minimum of $\Delta(s)$; adding H_C only *increases* the gap. There is no avoided crossing. The mixer is structurally incompatible with the landscape.

Correction: Grid $s^* = 0$ is not an “early avoided crossing.” For Grid on Ackley, Easom, Schwefel, and Eggholder, the gap minimum occurs exactly at $s = 0$ (confirmed numerically: gap increases monotonically from $s = 0$). This is *not* an early avoided crossing — it means the Grid Laplacian’s own normalised Fiedler value (≈ 0.010) is itself the bottleneck. No level crossing occurs during the interpolation; the grid simply lacks the connectivity to mix competing low-energy configurations. The mechanism is fundamentally different from a crossing, and this distinction matters for understanding when local mixers fail.

Table 4 records s^* for QMOA, Hypercube, and Grid across the mixer-sensitive functions, labelling the gap-closure mechanism for each.

Table 4: Critical point s^* for three key mixers. Red = $s^* = 0$ (Fiedler bottleneck).

Function	QMOA	Hyp.	Grid	Mechanism
Ackley	0.673	0.432	0.000	Grid: Fiedler bottleneck; QMOA: mid-path crossing
Easom	0.487	0.286	0.000	Grid: Fiedler bottleneck; QMOA near $s = 0.5$
Schwefel	0.990	1.000	0.000	Grid: Fiedler bottleneck
Eggholder	0.970	0.995	0.000	Grid: Fiedler bottleneck
Holder Table	0.990	0.568	0.191	Grid: early crossing; QMOA holds gap to $s \approx 1$
Rastrigin	0.990	0.899	0.834	All late; QMOA keeps gap largest throughout
Michalewicz	0.829	0.578	0.774	QMOA late-path; Grid mid-path collapse
Himmelblau	0.990	0.990	1.000	All near $s = 1$; H_C 4-fold degeneracy is bottleneck
Bukin N6	0.764	0.693	0.719	Mid-path crossing near $s \approx 0.7$ for all
Schaffer N2	0.965	0.995	0.141	Grid early collapse; QMOA/Hyp hold late
Griewank	0.965	0.925	0.533	Grid collapses at $s \approx 0.15$; QMOA holds to $s = 0.97$

6. Function Taxonomy: Five Behavioural Regimes

The 19 benchmark functions divide into five qualitatively distinct regimes.

Regime A — Mixer-blind (6 functions): Rosenbrock, Goldstein-Price, Matyas, Levy N13, Three Hump Camel, Cross-in-tray. All mixers achieve identical $\Delta_{\min}$; $s^* \approx 1$; bottleneck is H_C .

Regime B — QMOA wins (7 functions): Bukin N6, Rastrigin, Michalewicz, Holder Table, Ackley, Langerman, Easom. QMOA Complete achieves the largest or tied-largest $\Delta_{\min}$, with advantages of 1.1–10 $\times$ over Hypercube. Characteristic landscape: high ruggedness, small flat fraction, autocorrelation > 0.91 , axis-aligned multimodality.

Regime C — Hypercube competitive (3 functions): Schwefel, Schaffer N2, Eggholder. Hypercube marginally outperforms QMOA ($\leq 1.02\times$); these are effectively ties. Characteristic landscape: non-axis-aligned level-set geometry or near-checkerboard spatial autocorrelation.

Regime D — Grid wins (1 function: Himmelblau): Grid $\Delta_{\min} = 0.000173$ vs QMOA 0.000029 (6 $\times$) and Hypercube 0.000001 (173 $\times$). The four axis-symmetric global minima of Himmelblau create a near-degeneracy that QMOA’s row/column coupling reinforces, collapsing the gap. The Grid’s local structure is blind to the global four-fold symmetry.

Regime E — PCD variant wins (1 function: Griewank): The QMOA + Principal Complete Diagonals variant achieves the largest gap. Griewank’s product-of-cosines structure creates an interference pattern that aligns with the diagonal clique connectivity. This is an isolated result.

7. Why QMOA+PCD Consistently Underperforms

Adding complete-graph cliques along the main and anti-diagonal of the grid (the PCD variant) is harmful on 14 of 19 functions, despite increasing connectivity.

Mechanism. QMOA Complete has exactly three distinct normalised eigenvalues $\{0, \frac{1}{2}, 1\}$. Adding diagonal cliques introduces new eigenvalue clusters at intermediate values, breaking this clean three-level structure. During adiabatic interpolation, these new near-degenerate bundles can align with near-degenerate levels in H_C at intermediate s , creating additional avoided crossings that compress the gap.

Lesson: Spectral simplicity of H_Q is a *feature*, not a limitation. The fewer distinct eigenvalue clusters H_Q has, the fewer opportunities for destructive level crossings during interpolation.

Correction to earlier framing: The phrase “impact is strongly problem-dependent” understates the result. QMOA+PCD has average rank 6.5 vs 3.9 for plain QMOA across all 19 functions. The degradation is systematic, not incidental.

8. Landscape Feature Analysis and Predictive Rules

8.1 Quantitative Landscape Features

To connect landscape geometry to mixer performance, we compute five quantitative features for each function on a 64×64 grid:

Feature	Definition	Measures
Flat fraction	Fraction with $f_{\text{norm}} < 0.01$	Near-optimal basin size on discrete grid
Ruggedness	$\# \text{local minima} / (N-2)^2$	Density of competing basins
Separability	$\text{corr}(f, g(x) + h(y))$	Decomposability into x, y parts
Autocorrelation	$\text{corr}(f(i, j), \bar{f}_{\text{nbr}}(i, j))$	Spatial smoothness; < 0 = checkerboard
#Minima	Local minima count (8-neighbour comparison)	Absolute number of competing basins

Table 5: Landscape features and optimal mixer. Green = QMOA-regime indicator; orange = blind-regime indicator; red = Hypercube anomaly.

Function	#Min	Sep.	Rugg.	Flat	Autocorr.	Regime	Best
Rosenbrock	21	0.97	0.0055	0.2585	0.9984	blind	Any
Goldstein-Price	9	0.75	0.0023	0.5479	0.9795	blind	Any
Matyas	1	0.57	0.0003	0.0767	0.9829	blind	Any
Three Hump Camel	3	1.00	0.0008	0.4380	0.9995	blind	Any
McCormick	2	0.57	0.0005	0.0298	0.9840	blind	Any
Rastrigin	81	1.00	0.0211	0.0039	0.9922	QMOA	QMOA
Ackley	83	0.96	0.0216	0.0010	0.9984	QMOA	QMOA
Michalewicz	36	1.00	0.0094	0.0005	0.9992	QMOA	QMOA
Holder Table	42	0.90	0.0109	0.0010	0.9929	QMOA	QMOA
Langerman	338	0.24	0.0879	0.0010	0.9413	QMOA	QMOA
Easom	199	0.28	0.0518	0.0002	0.9105	QMOA	QMOA
Schaffer N2	363	0.11	0.0944	0.0005	−0.159	Hyp.	Hypercube
Eggholder	59	0.12	0.0153	0.0002	0.9430	Hyp.	Hypercube
Schwefel	49	1.00	0.0127	0.0002	0.9924	Hyp.	Hypercube
Himmelblau	4	0.85	0.0010	0.0366	0.9962	Grid	Grid
Griewank	6	0.31	0.0016	0.0078	0.9999	PCD	PCD

8.2 Three Discriminating Features

Flat fraction $\rightarrow$ **mixer-blind regime.** Functions with flat fraction > 0.05 are mixer-blind. Their discrete H_C has many near-degenerate low-energy states; the gap always collapses near $s = 1$ regardless of how amplitude is mixed at earlier stages. Average flat fraction: blind = 0.23, QMOA-wins = 0.001.

Ruggedness $\rightarrow$ mixer-sensitive regime. Higher ruggedness means more local minima $\rightarrow$ more mid-path level crossings $\rightarrow$ mixer topology determines which crossings are avoided. Average ruggedness: blind = 0.0025, QMOA-wins = 0.030, Hypercube = 0.041.

Spatial autocorrelation $\rightarrow$ Hypercube regime. All three Hypercube-competitive functions have autocorrelation < 0.95 ; all QMOA-wins functions have autocorrelation > 0.91 . Schaffer N2 (autocorr = -0.16) is the extreme case: a near-checkerboard landscape where the Hypercube’s ability to couple distant same-sign nodes is uniquely valuable.

Separability is not a reliable predictor alone. Schwefel (Hypercube wins) and Rastrigin (QMOA wins) both have separability = 1.00. What matters is whether the separability axis aligns with QMOA’s row/column decomposition. Rastrigin’s oscillations are axis-aligned; Schwefel’s are radially symmetric. The distinction cannot be captured by separability alone.

8.3 Empirical Predictive Rules

The following decision procedure correctly classifies 18/19 functions (Griewank remains an outlier):

Table 6: Feature-based mixer selection rules (empirical, 19 functions).

Condition	Regime	Recommended mixer
flat_frac > 0.05	Mixer-blind	Any
autocorr < 0.95	Hypercube	Hypercube Q10
rugg > 0.01 and sep > 0.80	QMOA	QMOA Complete
$N_{\min} \leq 5$ and sep > 0.80	Grid/Complete	Grid or Complete
(otherwise)	QMOA default	QMOA Complete

9. Summary of Numerical Observations

Across all benchmark functions, we observe that the spectral properties of QWOA are shaped jointly by the objective landscape and the structure of the quantum walk generator. Mixer-blind functions exhibit robust behaviour largely independent of graph choice. Mixer-sensitive functions display pronounced sensitivity to graph connectivity, with the critical point s^* falling into one of three mechanistically distinct modes.

Structured intermediate graphs such as the QMOA complete construction offer favourable trade-offs between global mixing and locality, motivating a systematic approach to ansatz design based on spectral and graph-theoretic considerations.

10. Conclusion and Outlook

In this work, we investigated how the choice of graph Laplacian defining the quantum walk Hamiltonian influences the spectral properties of QWOA. By analysing the full spectrum of $H(s) = (1 - s)H_Q + sH_C$ across nineteen benchmark functions and ten graph topologies, we demonstrated that graph structure plays a central role in shaping the adiabatic landscape of QWOA.

Our numerical results reveal a clear qualitative distinction between three regimes. For mixer-blind landscapes, the location and size of the minimum spectral gap are insensitive to the choice of quantum walk generator, with the bottleneck residing entirely in the near-degeneracy of H_C on the discrete grid. For mixer-sensitive rugged landscapes, the spectral gap exhibits a strong dependence on graph connectivity. A third, qualitatively new regime consists of functions where the gap minimum occurs at $s^* = 0$ for local mixers (the Fiedler bottleneck), indicating not

an avoided crossing during interpolation but a fundamental incompatibility between the mixer connectivity and the landscape structure.

Among the graph families studied, QMOA-inspired constructions exhibit the most consistently favourable spectral gap behaviour in the nontrivial regime, winning on 7 of 13 mixer-sensitive functions with advantages of up to $10\times$ over Hypercube. Three landscape features — flat fraction, ruggedness, and spatial autocorrelation — are empirically predictive of the optimal mixer class.

Importantly, QMOA is not universally optimal: Himmelblau provides a clear counterexample in which the Grid wins by $6\times$, because QMOA’s row/column structure reinforces the near-degeneracy introduced by the function’s four-fold axis-symmetric global minima. Similarly, adding diagonal cliques to QMOA (PCD variant) is systematically harmful rather than problem-dependent, because it destroys the clean $\{0, \frac{1}{2}, 1\}$ eigenvalue structure that underlies QMOA’s advantages.

Key takeaways.

1. QMOA Complete is the best general-purpose mixer: robust across diverse landscapes, catastrophic failure on none, wins 7/13 nontrivial cases.
2. The Fiedler bottleneck ($s^* = 0$) for Grid on oscillatory functions is a distinct failure mode, not an early avoided crossing.
3. Mixer-blind functions (large flat fraction) are insensitive to mixer choice; their apparent hardness is a property of H_C , not H_Q .
4. QMOA+PCD consistently degrades performance; spectral simplicity $\{0, \frac{1}{2}, 1\}$ is a structural advantage.
5. Three features (flat fraction, ruggedness, autocorrelation) suffice to predict the optimal mixer class on 18/19 benchmark functions.

Outlook. Future directions include: extending the analysis to larger system sizes and 3D cost functions; exploring adaptive or problem-dependent graph constructions informed by the predictive rules derived here; connecting adiabatic spectral features to the performance of gate-based and variational implementations of QWOA via Trotter-error analysis; and formalising the relationship between spatial autocorrelation and gap closure through spectral perturbation theory. More broadly, the framework developed here contributes to an emerging perspective in which structured quantum walk generators serve as a principled design tool for problem-aware quantum optimisation algorithms.

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